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Adaptive Incremental Broad Learning System Based on Interval Type-2 Fuzzy Set with Automatic Determination of Hyperparameters

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***Abstract*—The fuzzy broad learning system (FBLS) has received increasing attention due to its ability to quickly train from broad learning systems (BLS) and interpretability with fuzzy inference. However, the randomness of BLS brings instability to the training performance of the model, so the hyperparameters of the model are crucial for its performance. Currently, many FBLS use grid search to determine hyperparameters. However, grid search brings longer search time and the parameters obtained have randomness, which may not necessarily be the optimal hyperparameters. In response to these challenges, this paper proposes a fuzzy broad learning system with automatic determination of hyperparameters (ADHFBLS). We construct a novel FBLS based on the interval type-2 fuzzy set and design an incremental learning algorithm for rules and enhancement nodes to support rapid model expansion. Meanwhile, a heuristic hyperparameter automatic optimization algorithm is designed to overcome the randomness and long optimization time of grid search. Experiments have shown that ADHFBLS has higher accuracy and shorter model tuning time compared to some state-of-the-art models based on FBLS.**

***Index Terms*—Automatic determination of hyperparameters, fuzzy broad learning system, incremental learning, interval type-2 fuzzy set.**

I. Introduction

BROAD Learning System (BLS) is a novel neural network proposed by Chen and Liu [1]. Unlike deep neural networks, BLS does not require backpropagation for weight learning but instead learns parameters through matrix calculation, significantly reducing the training time and computational cost of the model. BLS extracts data features by constructing feature nodes and enhancement nodes, and can quickly expand the model through incremental learning algorithms to increase network complexity and adapt to new data.

Therefore, in recent years, there have been many works related to BLS. In [2], BLS was combined with CNN to solve the cross-domain emotion classification problem by integrating the strengths of CNN for feature extraction and BLS for efficient classifier training. Wang et al. embedded LSTM into BLS to achieve efficient processing of time series information, where BLS is used for feature extraction and LSTM is used to process time series [3]. In the review by Gong et al. in [4], they provided a detailed summary of some improvements made by BLS since its inception. This fully demonstrates the flexibility and efficiency of BLS, and more and more BLS-related works are being proposed, both in model improvement [5], [6] and application [7], [8], [9].

BLS has a simple structure, and fast training, and is expected to have good interpretability and robustness. Therefore, fuzzy inference has become a complementary fusion solution to further enhance the performance of BLS since fuzzy machine learning excels in uncertain data [10]. Fuzzy inference has a strong advantage in solving uncertain problems by simulating human decision-making processes using fuzzy sets and fuzzy rules. Since the computational process of fuzzy inference is essentially the inference of fuzzy rules, it has good interpretability. Among numerous fuzzy models, the Takagi-Sugeno (T-S) fuzzy model is an excellent fuzzy model that defines the fuzzy rule consequent as a linear function, has the characteristics of simple structure, strong interpretability, and good generalization, and is widely used in many fields [11], [12], [13]. Some works used more complex fuzzy sets to improve T-S fuzzy models to solve more complex problems. For example, type-2 fuzzy sets define membership as a new fuzzy set, which enhances the complexity of the model and its ability to solve uncertain problems. It has been validated by many works [14], [15], [16], [17], [18]. Therefore, using type-2 fuzzy sets to replace the fuzzy sets of T-S fuzzy models will improve the performance of T-S fuzzy models [19], [20].

Feng and Chen first proposed fuzzy BLS (FBLS) [21], which replaces the feature nodes of BLS with T-S fuzzy models, thereby improving the effectiveness of feature extraction. FBLS has proven that the fusion of the T-S fuzzy model and BLS is effective, and many related works have emerged. Peng et al. proposed the use of FBLS to solve fault

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detection caused by equipment aging and catalyst failure [22]. The T-S fuzzy model fully extracts the features of the original data and uses incremental learning algorithms to quickly reconstruct the network, effectively applying it to the penicillin fermentation platform and real industrial process. In [23], the authors proposed a stacked graph-regularized polynomial-based fuzzy broad learning system to address the challenges of low accuracy and efficiency in label enhancement models for label distribution learning. A type-2 fuzzy set-based BLS was proposed to enhance the rapid tracking response capability and robustness of the controller [24]. In addition, FBLS is widely used in fault diagnosis [25], obstacle-avoiding trajectory tracking [26], few-shot hyperspectral image classification [27], epilepsy detection [28], and so on.

However, FBLS is similar to BLS in that its model needs to randomly generate matrices during the construction process. Therefore, both FBLS and BLS have significant randomness. This randomness leads to unstable performance of the model, as the results of multiple training on the same model structure may deviate significantly, making it difficult to determine the hyperparameters of FBLS. The hyperparameters of FBLS typically include the number of fuzzy subsystems, the number of rules for each fuzzy subsystem, the number of enhancement nodes, etc. These hyperparameters have a crucial impact on the performance of the model. Many works use grid search to determine these hyperparameters [29], [30], [31], in order to select the best performing model. On the one hand, despite the short single training time of FBLS-based models, grid search still brings a long training time, especially in large-scale data. On the other hand, the randomness of FBLS determines that the hyperparameters obtained from grid search may not necessarily be the optimal hyperparameters. When training with the same hyperparameters again, the model may not achieve the same performance, and there may even be significant differences due to randomness. Therefore, eliminating the impact of hyperparameters in FBLS will improve model performance and training costs. However, similar work is still lacking at present.

To address these challenges, this paper proposes a fuzzy broad learning system with automatic determination of hyperparameters (ADHFBLS). Our contributions and innovations are summarized as follows:

1) We propose a novel FBLS based on an interval type-2 fuzzy set (IT2FS). We only retain one fuzzy subsystem and use IT2FS to enhance its feature extraction ability, thereby reducing the number of hyperparameters while enhancing the robustness and generalization of the model.
2) We design an incremental learning algorithm for enhancement nodes and fuzzy rules, which enables the model to further expand its size on the basis of already trained data, thereby enhancing its learning ability without the need for retraining.
3) Based on the algorithm proposed in (2), a heuristic algorithm for automatically determining hyperparameters is proposed and applied to the determination of the number of enhancement nodes and fuzzy rules in our proposed model. This method can quickly and adaptively adjust the hyperparameters of the model, and reduce the instability caused by the randomness of the model, thereby improving model performance and reducing training costs.

The rest of this paper is organized as follows: Section II introduces some related work. Section III provides a detailed description of the proposed model and algorithm, as well as theoretical proof. Section IV carries out experiments and the analysis of the results. Section V concludes this paper.

II. Related work

In recent years, many FBLS-based works have emerged to improve FBLS. Compact FBLS (CFBLS) was proposed to address the interpretability reduction caused by the presence of multiple subsystems in traditional FBLS [29]. CFBLS retains a fuzzy subsystem and uses the Interpretable Linguistic Fuzzy Rule (ILFR) to enhance the interpretability of the fuzzy subsystem. Similar works include [32] and [33], which respectively proposed a broad learning based dynamic fuzzy inference system (BL-DFIS) and a dynamic inference system in the fuzzy broad learning perspective (DFBLS). They all use ILFR to enhance the interpretability of fuzzy inference, where BL-DFIS introduces fuzzy inference into enhancement nodes, while feature nodes use Extreme Learning Machine Auto-Encoder (ELM-AE). DFBLS retains fuzzy subsystems on feature nodes while enhancement nodes use ILFR. It is worth mentioning that BL-DFIS and DFBLS have also proposed adaptive structure algorithms, as they both believe that enhancement nodes should be dynamically augmented through incremental learning to adapt to datasets of different sizes. Specifically, DFBLS proposes an incremental learning algorithm different from traditional BLS, which uses the Schur completion method for incremental computation of matrix pseudo inverse. However, the Schur completion method brings greater computational complexity and is not as concise as the Greville theorem, which is used in traditional BLS. In addition, both BL-DFIS and DFBLS consider the regularization parameter as one of the hyperparameters and use grid search to determine the optimal regularization parameter. This also indicates that regularization parameters have a significant impact on the performance of BLS-based models. Thus, [30] built a novel model of a fuzzy broad learning system based on accelerating amount to overcome the performance degradation caused by manually determining regularization parameters.

Unlike the aforementioned works that focus on interpretability, some works focus on improving the accuracy of FBLS. Another FBLS-based model, Extreme Fuzzy BLS (EFBLS), was proposed in [34]. EFBLS utilizes multiple FBLS to form a residual network to improve the accuracy of FBLS. This method constructs FBLS in the direction of depth, which has more advantages in accuracy but brings complexity to the network structure, especially in determining the hyperparameters of each layer of FBLS. Han et al. proposed T2F-BLS based on Type-2 fuzzy sets [35]. Type-2 fuzzy sets have higher complexity and higher dimensional uncertainty representation ability than traditional fuzzy sets. Therefore, T2F-BLS can effectively extract data features and improve accuracy. T2F-BLS also proposed an incremental learning algorithm aimed

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at adding new samples rather than enhancing the model's learning ability. Therefore, T2F-BLS also relies on grid search to determine hyperparameters.

TABLE I
COMPARISON OF RELATED WORK

Features	[29]	[32]	[33]	[30]	[34]	[35]	Ours
Fuzzy improving	✓	✓	✓			✓	✓
Pseudo inverse improving			✓	✓		✓	✓
Adaptability		✓	✓		✓		✓
None grid search							✓

Table I shows the comparison of existing work with our proposed model. Some of the existing works enhance the fuzzy representation by improving the fuzzy sets, some modify the computation of pseudo inverse to enhance the model performance, and some are equipped with adaptive structures. However, these works more or less use grid search to determine hyperparameters, and the hyperparameters that perform best on the testing set are considered optimal. Despite the fast training speed of BLS-based models, grid search will bring particularly long training time in large-scale datasets because the model needs to be retrained. Therefore, using incremental learning instead of retraining is an efficient method. However, the performance of BLS is not always improved during the incremental learning process [34]. Our proposed model will comprehensively consider these issues and overcome the shortcomings of existing work in grid search dependency and hyperparameter determination.

III. Fuzzy Broad Learning System with Automatic Determination of Hyperparameters

This section provides a detailed description of ADHFBLS. Firstly, we briefly describe the original FBLS. Then, the IT2FS-based FBLS proposed in this paper is introduced. Following that, the incremental learning algorithm is elaborated in detail. Finally, we introduce the automatic hyperparameter tuning algorithm proposed in this paper.

A. Structure of Original FBLS

FBLS is a model developed based on BLS, which not only has the characteristics of fast training but also further enhances the ability to handle uncertain problems and interpretability [21]. Specifically, FBLS uses the T-S fuzzy model to replace the feature mapping nodes of BLS, fuzzifying the data. The fuzzified data is then processed through the calculation of the enhancement nodes and finally multiplied by the weight matrix solved by ridge regression to obtain the output, thereby solving tasks such as classification and regression.

We assume that the training sample set size is N and the number of features is M, denoted as $\boldsymbol{X} = (\boldsymbol{x}_1, \boldsymbol{x}_2, ..., \boldsymbol{x}_N)^T \in \mathbb{R}^{N\times M}$, where $\boldsymbol{x}_s = (x_{s1}, x_{s2}, ..., x_{sM})$. The label of the sample is denoted as $\boldsymbol{Y} = (\boldsymbol{y}_1, \boldsymbol{y}_2, ..., \boldsymbol{y}_N) \in \mathbb{R}^{N\times C}$, where C is the label dimension. FBLS has n fuzzy subsystems and the number of rules for each fuzzy subsystem is K_i, which is denoted in the following form for the kth rule of the ith system:

$$\text{If } x_{s1} \text{ is } A^i_{k1} \text{ and } x_{s2} \text{ is } A^i_{k2} \text{ ... and } x_{sM} \text{ is } A^i_{kM}$$
$$\text{Then } z^i_{sk} = \sum_{t=1}^{M} \alpha^i_{kt} x_{st} \tag{1}$$

where A^i_{kt} is the membership function, α^i_{kt} is the rule's consequent coefficient of the T-S fuzzy model, which is randomly generated from $[0, 1]$, $i = 1, 2, ..., n, k = 1, 2, ..., K_i, t = 1, 2, ..., M$. FBLS uses Gaussian membership function, so A^i_{kt} is expressed as follows:

$$A^i_{kt}(x_{st}) = \exp(-(\frac{x_{st} - c^i_{kt}}{\sigma^i_{kt}})^2) \tag{2}$$

where σ^i_{kt} is the fixed width and c^i_{kt} is the center determined by the k-means algorithm. The weighted firing strength for each rule is

$$\omega^i_{sk} = \frac{\prod_{t=1}^{M} A^i_{kt}(x_{st})}{\sum_{k'=1}^{K_i}\prod_{t=1}^{M} A^i_{k't}(x_{st})}, k = 1, 2, ..., K_i \tag{3}$$

From this, the output of the ith fuzzy subsystem for the sth training sample is

$$\boldsymbol{Z}^i_s = (\omega^i_{s1} z^i_{s1}, \omega^i_{s2} z^i_{s2}, ..., \omega^i_{sK_i} z^i_{sK_i}) \tag{4}$$

and the output of the ith fuzzy subsystem for all training samples is

$$\boldsymbol{Z}^i = (\boldsymbol{Z}^i_1, \boldsymbol{Z}^i_2, ..., \boldsymbol{Z}^i_N)^T \in \mathbb{R}^{N\times K_i} \tag{5}$$

Thus, the whole output of all fuzzy subsystems is expressed as

$$\boldsymbol{Z} = (\boldsymbol{Z}^1, \boldsymbol{Z}^2, ..., \boldsymbol{Z}^n) \in \mathbb{R}^{N\times \sum_{i=1}^{n} K_i} \tag{6}$$

$\boldsymbol{Z}$ will be fed into the enhancement node for further computation. We assume that the number of enhancement nodes is m, and the output of the jth enhancement node is denoted as

$$\boldsymbol{H}_j = \xi(\boldsymbol{Z}\boldsymbol{W}_j + \beta_j) \tag{7}$$

where $\xi(\cdot)$ is the Sigmoid activation function, $\boldsymbol{W}_j$ and β_j are the randomly generated weight matrix and bias item, $j = 1, 2, ..., m$. The output of all enhancement nodes is represented as

$$\boldsymbol{H} = (\boldsymbol{H}_1, \boldsymbol{H}_2, ..., \boldsymbol{H}_m) \in \mathbb{R}^{N\times m} \tag{8}$$

$\boldsymbol{Z}$ and $\boldsymbol{H}$ are connected and multiplied by the weight matrix $\boldsymbol{W}$ to get the final output:

$$\hat{\boldsymbol{Y}} = (\boldsymbol{Z}, \boldsymbol{H})\boldsymbol{W} \tag{9}$$

where $\boldsymbol{W}$ is the weight matrix to be learned and is computed by the following equation:

$$\boldsymbol{W} = (\boldsymbol{Z}, \boldsymbol{H})^{\dagger}\boldsymbol{Y} \tag{10}$$

where $(\boldsymbol{Z}, \boldsymbol{H})^{\dagger}$ is the pseudo inverse of $(\boldsymbol{Z}, \boldsymbol{H})$ and is calculated through ridge regression.

FBLS has been proven to have good performance in classification and regression problems. However, FBLS uses grid search to optimize the number of rules, fuzzy subsystems, and enhancement nodes, which can result in significant time

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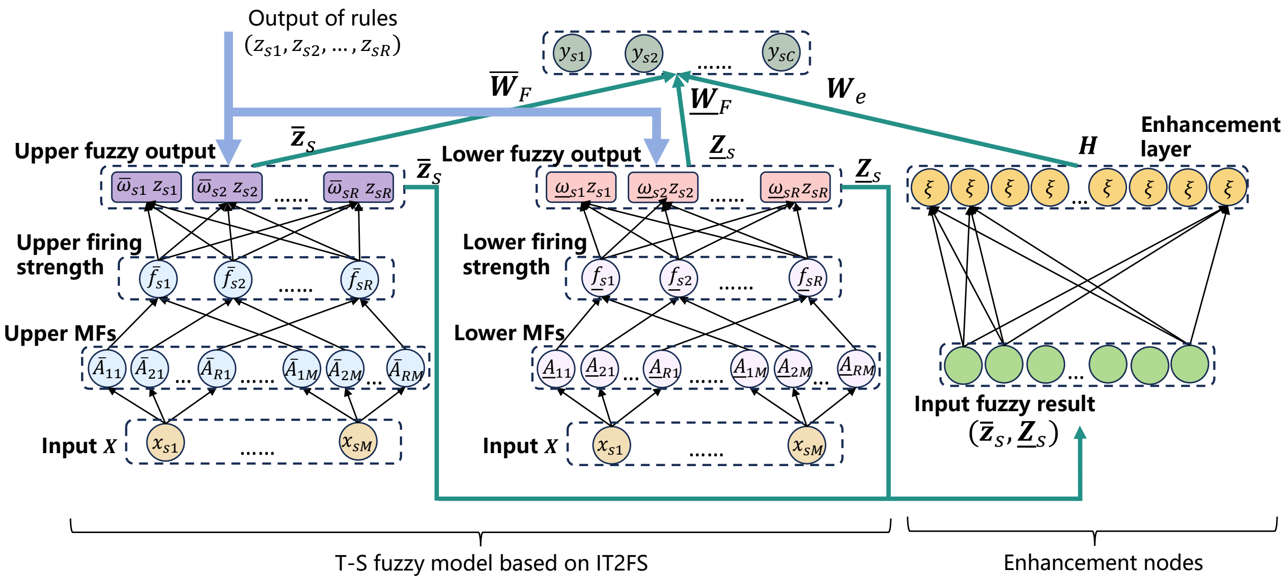

Fig. 1. The basic structure of ADHFBLS. In this diagram, the membership function is abbreviated as MF.

consumption on large-scale datasets. Moreover, due to the randomness of the model, using the best hyperparameters found for repeated training may also result in performance not reaching the original level.

B. Basic Structure of ADHFBLS

In the original FBLS, multiple fuzzy subsystems work together to fuzzify the data. To reduce the dimensionality of hyperparameters, we only use one fuzzy subsystem based on IT2FS. Compared to traditional fuzzy sets, IT2FS has more complicated fuzzification capabilities and more efficient data feature extraction due to its more abundant fuzzy representation. This characteristic compensates for the decrease in fuzzy representation capability caused by our use of a single fuzzy subsystem and improves the accuracy of the model.

TABLE II
DECLARATION OF KEY VARIABLES IN ADHFBLS

Valuable	Symbol
N, M	Number of training samples and features
R, m	Rule number and enhancement node number
η	Enhancement rate
$\bar{A}, \underline{A}$	Upper MF and lower MF
$\bar{c}, \underline{c}, \sigma$	Upper center, lower center, and width
α, z	Rule's consequent coefficient and output of rule
$\bar{f}, \underline{f}, \bar{\omega}, \underline{\omega}$	Upper and lower firing strength and weighted forms
$\bar{\boldsymbol{Z}}, \underline{\boldsymbol{Z}}$	Upper and lower fuzzy output
$\boldsymbol{H}$	Enhancement node output
$\boldsymbol{D}$	Connection of $\bar{\boldsymbol{Z}}, \underline{\boldsymbol{Z}}, \boldsymbol{H}$
$\boldsymbol{W}$	Weight matrix

The basic structure of ADHFBLS is shown in Fig. 1. Some key variables are declared as shown in Table II. In ADHFBLS, the data first passes through a T-S fuzzy model based on IT2FS. Assuming there are a total of R rules. The membership function is defined by the following equations [36]:

$$\bar{A}_{rt}(x_{st}) = \begin{cases} \exp(-\frac{(x_{st}-\underline{c}_{rt})^2}{2\sigma_{rt}^2}), x_{st} \le \underline{c}_{rt} \\ 1, \quad \underline{c}_{rt} < x_{st} \le \bar{c}_{st} \\ \exp(-\frac{(x_{st}-\bar{c}_{rt})^2}{2\sigma_{rt}^2}), x_{st} > \bar{c}_{rt} \end{cases} \tag{11}$$

$$\underline{A}_{rt}(x_{st}) = \begin{cases} \exp(-\frac{(x_{st}-\bar{c}_{rt})^2}{2\sigma_{rt}^2}), x_{st} \le \frac{\underline{c}_{rt}+\bar{c}_{rt}}{2} \\ \exp(-\frac{(x_{st}-\underline{c}_{rt})^2}{2\sigma_{rt}^2}), x_{st} > \frac{\underline{c}_{rt}+\bar{c}_{rt}}{2} \end{cases} \tag{12}$$

where $\bar{A}_{kt}(\cdot)$ and $\underline{A}_{kt}(\cdot)$ are the upper MFs and lower MFs, $r = 1, 2, ..., R$, $t = 1, 2, ..., M$. $\bar{c}_{rt}$ and $\underline{c}_{rt}$ are the upper centers and lower centers, respectively, uniformly sampled within $[0.5, 1]$ and $[0, 0.5]$. σ_{rt} is the width, which is set to 0.5 in this paper. Therefore, the upper and lower firing strengths are represented as

$$\bar{f}_{sr} = \prod_{t=1}^{M} \bar{A}_{rt}(x_{st}) \tag{13}$$

$$\underline{f}_{sr} = \prod_{t=1}^{M} \underline{A}_{rt}(x_{st}) \tag{14}$$

The posterior output of the rule is calculated using Eq. (1), so the upper and lower fuzzy outputs of the sth training sample are represented as

$$\begin{aligned} \bar{\boldsymbol{Z}}_s &= (\bar{\omega}_{s1}z_{s1}, \bar{\omega}_{s2}z_{s2}, ..., \bar{\omega}_{sR}z_{sR}) \\ &= \frac{1}{\sum_{r=1}^{R}\bar{f}_{sr}}(\bar{f}_{s1}z_{s1}, \bar{f}_{s2}z_{s2}, ..., \bar{f}_{sR}z_{sR}) \end{aligned} \tag{15}$$

$$\begin{aligned} \underline{\boldsymbol{Z}}_s &= (\underline{\omega}_{s1}z_{s1}, \underline{\omega}_{s2}z_{s2}, ..., \underline{\omega}_{sR}z_{sR}) \\ &= \frac{1}{\sum_{r=1}^{R}\underline{f}_{sr}}(\underline{f}_{s1}z_{s1}, \underline{f}_{s2}z_{s2}, ..., \underline{f}_{sR}z_{sR}) \end{aligned} \tag{16}$$

where $\bar{\omega}_{s1}, \bar{\omega}_{s2}, ..., \bar{\omega}_{sR}$ is the upper weighted firing strength, and $\underline{\omega}_{s1}, \underline{\omega}_{s2}, ..., \underline{\omega}_{sR}$ is the lower weighted firing strength. The

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upper and lower fuzzy outputs of all samples are represented as

$$\bar{\boldsymbol{Z}} = (\bar{\boldsymbol{Z}}_1, \bar{\boldsymbol{Z}}_2, ..., \bar{\boldsymbol{Z}}_s)^T \in \mathbb{R}^{N\times R} \tag{17}$$

$$\underline{\boldsymbol{Z}} = (\underline{\boldsymbol{Z}}_1, \underline{\boldsymbol{Z}}_2, ..., \underline{\boldsymbol{Z}}_s)^T \in \mathbb{R}^{N\times R} \tag{18}$$

$\bar{\boldsymbol{Z}}$ and $\underline{\boldsymbol{Z}}$ are concatenated and input into the enhancement node, calculated using Eqs. (7), (8), where $\boldsymbol{Z} = (\bar{\boldsymbol{Z}}, \underline{\boldsymbol{Z}})$. It should be noted that in the basic ADHFBLS, which has not yet undergone incremental learning, the number of enhancement nodes needs to be an integer multiple, which we define as enhancement rate η, of the number of rules, satisfying the following equations:

$$m = \eta \times R \tag{19}$$

where $\eta, R \in \mathbb{N}^{+}$. The final output of ADHFBLS is

$$\begin{aligned} \hat{\boldsymbol{Y}} &= \bar{\boldsymbol{Z}}\bar{\boldsymbol{W}}_F + \underline{\boldsymbol{Z}}\underline{\boldsymbol{W}}_F + \boldsymbol{H}\boldsymbol{W}_e \\ &= (\bar{\boldsymbol{Z}}, \underline{\boldsymbol{Z}}, \boldsymbol{H}) \begin{pmatrix} \bar{\boldsymbol{W}}_F \\ \underline{\boldsymbol{W}}_F \\ \boldsymbol{W}_e \end{pmatrix} \\ &= \boldsymbol{D}\boldsymbol{W} \end{aligned} \tag{20}$$

where $\bar{\boldsymbol{W}}_F$, $\underline{\boldsymbol{W}}_F$ and $\boldsymbol{W}_e$ are the weight matrices for the upper fuzzy output, lower fuzzy output, and enhancement node output, respectively, and $\boldsymbol{D} = (\bar{\boldsymbol{Z}}, \underline{\boldsymbol{Z}}, \boldsymbol{H})$. $\boldsymbol{W}$ is calculated using Eq. (10), but the difference is that we use singular value decomposition (SVD) instead of ridge regression. This is because ridge regression has a regularization parameter, which is also considered a hyperparameter that requires grid search in some works. Therefore, SVD is used to avoid the introduction of hyperparameters and the accumulation of errors in subsequent incremental learning, which can lead to a decrease in model performance.

C. Incremental Learning Algorithm for ADHFBLS

From the basic model of ADHFBLS, it can be seen that it only has two hyperparameters: the number of rules R and the enhancement rate η. Therefore, even if the hyperparameters of the model are determined through grid search, it is still more efficient than existing FBLS-based models. However, grid search requires the network to be retrained, so we hope to achieve incremental learning to avoid retraining the network while adjusting hyperparameters, thereby further reducing the cost of hyperparameter adjustment.

ADHFBLS provides two ways of incremental learning, namely the increment of enhancement nodes and the increment of fuzzy rules. In this section, we will provide a detailed introduction to their incremental learning process.

1) Incremental Learning of Enhancement Nodes: The incremental learning of enhancement nodes is the same as that of traditional BLS, and most existing work extends the model in this way. Specifically, we assume that ADHFBLS originally has $m = \eta \times R$ enhancement nodes, and then we need to add $\Delta m = \Delta\eta \times R$ enhancement nodes to increase the complexity of the model. During this process, $(\bar{\boldsymbol{Z}}, \underline{\boldsymbol{Z}})$ do not change, only $\boldsymbol{H}$ add elements, which are represented as

$$\begin{aligned} \boldsymbol{H}^{*} &= (\boldsymbol{H}_1, \boldsymbol{H}_2, ..., \boldsymbol{H}_m | \boldsymbol{H}^{ex}_1, \boldsymbol{H}^{ex}_2, ..., \boldsymbol{H}^{ex}_{\Delta m}) \\ &= (\boldsymbol{H}|\boldsymbol{H}^{ex}) \end{aligned} \tag{21}$$

where $\boldsymbol{H}^{*}$ represents the enhancement nodes output after increment, and $\boldsymbol{H}^{ex}$ is obtained by randomly generating more $\boldsymbol{W}_j$ and β_j, and using Eq. (7). $\boldsymbol{D}$ has been updated to $(\boldsymbol{D}|\boldsymbol{H}^{ex})$, denoted as $\boldsymbol{D}^{*}$. Noting that the expanded $\boldsymbol{W}$ is $\boldsymbol{W}^{*}$, then

$$\boldsymbol{W}^{*} = (\boldsymbol{D}^{*})^{\dagger}\boldsymbol{Y} = (\boldsymbol{D}|\boldsymbol{H}^{ex})^{\dagger}\boldsymbol{Y} \tag{22}$$

We can incrementally use the calculated $\boldsymbol{D}^{\dagger}$ to calculate $(\boldsymbol{D}^{*})^{\dagger}$, using the following equations:

$$(\boldsymbol{D}^{*})^{\dagger} = (\boldsymbol{D}|\boldsymbol{H}^{ex})^{\dagger} = \begin{bmatrix} \boldsymbol{D}^{\dagger} - \boldsymbol{U}\boldsymbol{B}^{T} \\ \boldsymbol{B}^{T} \end{bmatrix} \tag{23}$$

where $\boldsymbol{U} = \boldsymbol{D}^{\dagger}\boldsymbol{H}^{ex}$,

$$\boldsymbol{B}^{T} = \begin{cases} \boldsymbol{C}^{\dagger} & \text{if } \boldsymbol{C} \neq \boldsymbol{0} \\ (\boldsymbol{I} + \boldsymbol{U}^{T}\boldsymbol{U})^{-1}\boldsymbol{U}^{T}\boldsymbol{D}^{\dagger} & \text{if } \boldsymbol{C} = \boldsymbol{0} \end{cases}$$

and $\boldsymbol{C} = \boldsymbol{H}^{ex} - \boldsymbol{D}\boldsymbol{U}$.

Enhancement nodes perform nonlinear transformations on fuzzy data to further obtain information from the fuzzy data and improve the accuracy of the model. By incremental learning of enhancement nodes, the expansion of the model does not require retraining and only requires further computation on the original weight matrix, thereby reducing computation time. However, if the fuzzification of the data is not sufficient, it will hinder the improvement of enhancement nodes' performance. Therefore, incremental learning of fuzzy rules is also required in the expansion of the model.

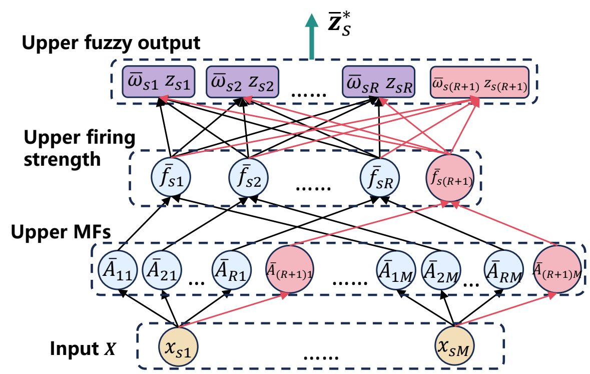

Fig. 2. Schematic diagram of adding rules to the upper part of the fuzzy model. In this diagram, a rule has been added, where the red components represent the added computing units and outputs.

2) Incremental Learning of Fuzzy Rules: The incremental learning of fuzzy rules is a method we innovatively propose. By increasing the number of fuzzy rules, the model enhances its ability to fuzzify data. Similar to the incremental learning of enhancement nodes, the incremental learning of fuzzy rules does not require retraining the model, thereby reducing training time. We assume that ADHFBLS originally has R rules and the number of rules to be added is ΔR. For the added rules, their upper and lower centers, as well as the coefficients of the rules' consequent, are also randomly generated. The schematic diagram of adding model rules is shown in Fig. 2. From the graph, it can be seen that the increase in rules brings more changes compared to the increase in enhancement nodes. After the increase in the number of rules, $\bar{\boldsymbol{Z}}$ and $\underline{\boldsymbol{Z}}$

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change, and correspondingly, $\boldsymbol{H}$ also change. To this end, we use approximation methods to transform these changes into a form similar to Eq. (22).

Specifically, for the sth training sample, its upper fuzzy output is updated to

$$\begin{aligned}\bar{\boldsymbol{Z}}_s^* &= \frac{1}{\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}}(\bar{f}_{s1}z_{s1},...,\bar{f}_{sR}z_{sR},...,\bar{f}_{s(R+\Delta R)}z_{s(R+\Delta R)}) \\ &= (\frac{\sum_{r=1}^{R}\bar{f}_{sr}}{\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}}\bar{\boldsymbol{Z}}_s|\bar{\boldsymbol{Z}}_s^{ex}) \end{aligned} \tag{24}$$

where

$$\bar{\boldsymbol{Z}}_s^{ex} = \frac{1}{\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}}(\bar{f}_{s(R+1)}z_{s(R+1)},...,\bar{f}_{s(R+\Delta R)}z_{s(R+\Delta R)})$$

The upper fuzzy output of all samples is

$$\bar{\boldsymbol{Z}}^* = (\bar{\boldsymbol{\Lambda}}\bar{\boldsymbol{Z}}|\bar{\boldsymbol{Z}}^{ex}) \tag{25}$$

where $\bar{\boldsymbol{Z}}^{ex} = (\bar{\boldsymbol{Z}}_1^{ex},...,\bar{\boldsymbol{Z}}_N^{ex})^T$ and $\bar{\boldsymbol{\Lambda}} = \text{diag}\{\frac{\sum_{r=1}^{R}\bar{f}_{1r}}{\sum_{r=1}^{R+\Delta R}\bar{f}_{1r}},...,\frac{\sum_{r=1}^{R}\bar{f}_{Nr}}{\sum_{r=1}^{R+\Delta R}\bar{f}_{Nr}}\} \in \mathbb{R}^{N\times N}$. Since the upper center of each rule is sampled from the same distribution, the following equation is satisfied:

$$\mathbb{E}[\bar{f}_{sr}] = \mathbb{E}[\bar{f}_{sp}], p = 1,2,...,R+\Delta R \tag{26}$$

Therefore, we have the following estimate:

$$\begin{aligned}\frac{\sum_{r=1}^{R}\bar{f}_{sr}}{\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}} &\approx \frac{\mathbb{E}[\sum_{r=1}^{R}\bar{f}_{sr}]}{\mathbb{E}[\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}]} = \frac{\sum_{r=1}^{R}\mathbb{E}[\bar{f}_{sr}]}{\sum_{r=1}^{R+\Delta R}\mathbb{E}[\bar{f}_{sr}]} \\ &= \frac{R}{R+\Delta R}\end{aligned} \tag{27}$$

Based on Eq. (27), we estimate $\bar{\boldsymbol{Z}}^*$ as

$$\bar{\boldsymbol{Z}}^* \approx (\frac{R}{R+\Delta R}\bar{\boldsymbol{Z}}|\bar{\boldsymbol{Z}}^{ex}) \tag{28}$$

Similarly, we can obtain the lower fuzzy output of all samples

$$\underline{\boldsymbol{Z}}^* \approx (\frac{R}{R+\Delta R}\underline{\boldsymbol{Z}}|\underline{\boldsymbol{Z}}^{ex}) \tag{29}$$

Next, we will consider the changes in the output of enhancement nodes. We provide the following theorem and prove it.

Theorem 1. *After adding rules, the output of enhancement nodes remains approximately unchanged, denoted as* $\boldsymbol{H}^* \approx \boldsymbol{H}$.

Proof. We take the calculation result of the sth sample at the jth enhancement node, denoted as

$$h_{sj} = \xi((\bar{\boldsymbol{Z}}_s,\underline{\boldsymbol{Z}}_s)\boldsymbol{W}_j + \beta_j) \tag{30}$$

$$h_{sj}^* = \xi((\bar{\boldsymbol{Z}}_s^*,\underline{\boldsymbol{Z}}_s^*)\boldsymbol{W}_j^* + \beta_j) \tag{31}$$

where h_{sj} and h_{sj}^* are the outputs before and after adding the rule, respectively. We consider the values before activation, and their difference is calculated as

$$\begin{aligned}\varepsilon &= (\bar{\boldsymbol{Z}}_s,\underline{\boldsymbol{Z}}_s)\boldsymbol{W}_j + \beta_j - (\bar{\boldsymbol{Z}}_s^*,\underline{\boldsymbol{Z}}_s^*)\boldsymbol{W}_j^* - \beta_j \\ &= \underbrace{\frac{\sum_{r=1}^{R}\bar{f}_{sr}z_{sr}\bar{w}_{rj}}{\sum_{r=1}^{R}\bar{f}_{sr}} - \frac{\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}z_{sr}\bar{w}_{rj}}{\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}}}_{\bar{\varepsilon}} \\ &+ \underbrace{\frac{\sum_{r=1}^{R}\underline{f}_{sr}z_{sr}\underline{w}_{rj}}{\sum_{r=1}^{R}\underline{f}_{sr}} - \frac{\sum_{r=1}^{R+\Delta R}\underline{f}_{sr}z_{sr}\underline{w}_{rj}}{\sum_{r=1}^{R+\Delta R}\underline{f}_{sr}}}_{\underline{\varepsilon}}\end{aligned} \tag{32}$$

We calculate the result of $\bar{\varepsilon}$, and the calculation process of $\underline{\varepsilon}$ is consistent with it. Further

$$\begin{aligned}\bar{\varepsilon} &= (\frac{1}{\sum_{r=1}^{R}\bar{f}_{sr}} - \frac{1}{\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}})\sum_{r=1}^{R}\bar{f}_{sr}z_{sr}\bar{w}_{rj} \\ &- \frac{\sum_{r=R+1}^{R+\Delta R}\bar{f}_{sr}z_{sr}\bar{w}_{rj}}{\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}} \\ &= \frac{1}{\sum_{r=1}^{R}\bar{f}_{sr}\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}}(\sum_{r=1}^{R}\bar{f}_{sr}z_{sr}\bar{w}_{rj}\sum_{r=R+1}^{R+\Delta R}\bar{f}_{sr} \\ &- \sum_{r=R+1}^{R+\Delta R}\bar{f}_{sr}z_{sr}\bar{w}_{rj}\sum_{r=1}^{R}\bar{f}_{sr}) \\ &= \frac{\sum_{r=1}^{R}(\bar{f}_{sr}\sum_{t=R+1}^{R+\Delta R}\bar{f}_{st}(z_{sr}\bar{w}_{rj} - z_{st}\bar{w}_{tj}))}{\sum_{r=1}^{R}\bar{f}_{sr}\sum_{r=1}^{R+\Delta R}\bar{f}_{sr}}\end{aligned} \tag{33}$$

Since both $\boldsymbol{W}_j$ and the consequent of the rule are randomly sampled from the same distribution, the following equation is satisfied:

$$\mathbb{E}[z_{sr}\bar{w}_{rj}] = \mathbb{E}[z_{st}\bar{w}_{tj}], r,t = 1,2,...,R+\Delta R \tag{34}$$

Therefore, we can draw the following conclusion

$$\mathbb{E}[\bar{\varepsilon}] = 0 \tag{35}$$

$\underline{\varepsilon}$ has the same property, from which we get

$$\varepsilon = \bar{\varepsilon} + \underline{\varepsilon} \approx 0 \tag{36}$$

At this point, we have completed the proof of the theorem by taking over all s and j. □

If we denote the fuzzy output and enhancement nodes output as $\boldsymbol{D}^*$, then

$$\begin{aligned}\boldsymbol{D}^* &= (\bar{\boldsymbol{Z}}^*,\underline{\boldsymbol{Z}}^*,\boldsymbol{H}^*) \triangleq (\frac{R}{R+\Delta R}\bar{\boldsymbol{Z}},\frac{R}{R+\Delta R}\underline{\boldsymbol{Z}},\boldsymbol{H}|\bar{\boldsymbol{Z}}^{ex},\underline{\boldsymbol{Z}}^{ex}) \\ &= \frac{R}{R+\Delta R}(\bar{\boldsymbol{Z}},\underline{\boldsymbol{Z}},\frac{R+\Delta R}{R}\boldsymbol{H}|\frac{R+\Delta R}{R}\bar{\boldsymbol{Z}}^{ex},\frac{R+\Delta R}{R}\underline{\boldsymbol{Z}}^{ex})\end{aligned} \tag{37}$$

For $\frac{R+\Delta R}{R}\boldsymbol{H}$, it indicates that each element of $\boldsymbol{H}$ has increased by $\frac{\Delta R}{R}$ times. Since $\boldsymbol{H}$ is multiplied by weights and summed, we can use more enhancement nodes to eliminate the influence of multiples. Hence, the number of enhancement nodes increases to

$$\begin{aligned}m^* &= \frac{R+\Delta R}{R}m = \frac{R+\Delta R}{R}\eta R = \eta R + \eta\Delta R \\ &= m + \eta\Delta R\end{aligned} \tag{38}$$

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The newly added enhancement nodes are represented as $\boldsymbol{H}^{ex} \in \mathbb{N}^{N\times\eta\Delta R}$. Thus, $\boldsymbol{D}^*$ can be represented in the following form:

$$\begin{aligned}\boldsymbol{D}^* &= \frac{R}{R+\Delta R}(\bar{\boldsymbol{Z}}, \underline{\boldsymbol{Z}}, \boldsymbol{H}|\frac{R+\Delta R}{R}\bar{\boldsymbol{Z}}^{ex}, \frac{R+\Delta R}{R}\underline{\boldsymbol{Z}}^{ex}, \boldsymbol{H}^{ex}) \\ &= \frac{R}{R+\Delta R}(\boldsymbol{D}|\frac{R+\Delta R}{R}\bar{\boldsymbol{Z}}^{ex}, \frac{R+\Delta R}{R}\underline{\boldsymbol{Z}}^{ex}, \boldsymbol{H}^{ex})\end{aligned} \tag{39}$$

In this way, $\boldsymbol{D}^*$ will be able to incrementally use following equation:

$$(\boldsymbol{D}^*)^\dagger = \frac{R+\Delta R}{R}(\boldsymbol{D}|\frac{R+\Delta R}{R}\bar{\boldsymbol{Z}}^{ex}, \frac{R+\Delta R}{R}\underline{\boldsymbol{Z}}^{ex}, \boldsymbol{H}^{ex})^\dagger \tag{40}$$

and Eq. (23) on the basis of $\boldsymbol{D}^\dagger$.

For incremental learning of fuzzy rules, we have the following remarks: 1) Incremental learning of fuzzy rules is more complex than incremental learning of enhancement nodes, and there is no precise method for this type of incremental learning. We innovatively use approximate methods for it. This method does not necessarily mean that it will cause errors in the performance of the model, as many parameters of ADHFBLS are randomly generated. When we use Eq. (39) for estimation, we can assume that the previously randomly generated rule consequent coefficients and matrices have been modified, as if they were originally randomly generated like this. Our design ensures that this change is minimal, so the previously learned $\boldsymbol{W}$ is still effective; 2) We move the newly generated data to the rightmost part of the matrix, such as the transformation of Eq. (37), to use Eq. (23) to achieve incremental computation of pseudo inverse, and this transformation does not affect the performance of the model; 3) The incremental learning of fuzzy rules requires the number of enhancement nodes to be an integer multiple of the rules, and Eq. (38) indicates the reason for this.

D. Automatic Determination of Hyperparameters

In the previous section, we implement two incremental learning algorithms for ADHFBLS, for which we are able to increase the size of the model faster thus matching the size of the training data. For most BLS systems, proper hyperparameters are crucial, since too small a network size is insufficient in terms of performance, while too large a network size will make the model less generalizable. For this reason, we design an algorithm for automatic hyperparameter determination to find the optimal hyperparameters for ADHFBLS.

To evaluate the generalization of the model, the training dataset is generally divided into a portion as the validation dataset. The validation dataset is used to determine the setting of hyperparameters so that the model has better generalization on the test dataset and we define it as $\check{\boldsymbol{X}}$. We define the hyperparameters of ADHFBLS as $(R, \eta R)$, representing R rules and ηR enhancement nodes. We can choose to add δ rules or $\eta\delta$ enhancement nodes to ADHFBLS, represented as $(R+\delta, \eta R+\eta\delta)$ and $(R, \eta R+\eta\delta)$. We use the one that can improve model performance the most on $\check{\boldsymbol{X}}$, and let $\delta \leftarrow 2\times\delta$, which can make the increase of hyperparameters faster. If the model performance does not improve after incremental

Algorithm 1: Training Algorithm for ADHFBLS.

Data: $\boldsymbol{X}$, $\check{\boldsymbol{X}}$ and their labels, initial rule number R and enhancement rate η, τ and t_{max}.
Result: trained ADHFBLS θ.

```
δ = 1;
Train the initial ADHFBLS θ on X using
  hyperparameter (R, ηR) according to Eq. (11)-Eq.
  (20), and get the accuracy acc on X̌ and its labels;
t = 0;
while t < t_max do
    acc_rule = 0, acc_enhance = 0;
    for i = 1 to τ do
        Add δ rules to θ (along with ηδ enhancement
          nodes) and update W using the dataset X to
          obtain the model θ̂*_r according to Eq. (24),
          Eq. (25), Eq. (38)-Eq. (40);
        Test θ̂*_r using the dataset X̌ and its labels to
          obtain accuracy;
        Update the optimal accuracy acc_rule and the
          optimal model θ*_r;
    for i = 1 to τ do
        Add ηδ enhancement nodes to θ and update W
          using the dataset X to obtain the model θ̂*_e
          according to Eq. (21)- Eq. (23);
        Test θ̂*_e using the dataset X̌ and its labels to
          obtain accuracy;
        Update the optimal accuracy acc_enhance and
          the optimal model θ*_e;
    if acc_rule > acc_enhance and acc_rule ≥ acc
      then
        θ ← θ*_r;
        δ ← 2 × δ;
    else if acc_rule ≤ acc_enhance and
      acc_enhance ≥ acc then
        θ ← θ*_e;
        δ ← 2 × δ;
    else
        if δ == 1 then
            break;
        δ ← 1;
    t ← t + 1;
return θ
```

TABLE III
DETAILS OF NINE DATASETS IN UCI

Datasets	No. of samples	No. of features	No. of classes
Diabetes	768	8	2
Vehicle	846	18	4
Contraceptive	1473	9	3
Penbased	7494	16	10
Nursery	12960	8	5
Online Shoppers	12330	17	2
Adult	32561	14	2
Bank Marketing	41188	20	2
Connect-4	67557	42	3

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TABLE IV
MODEL ACCURACY OF ADHFBLS AND BASELINE MODELS

Datasets	**FBLS**	**BL-DFIS**	**DFBLS**	**EFBLS**	**T2F-BLS**	**IFBLS**	**ADHFBLS-grid**	**ADHFBLS**
Diabetes	0.7749	0.7835	0.7706	0.7792	0.7879	0.7835	0.7835	**0.7965**
Vehicle	0.8425	0.8425	0.8465	**0.8622**	0.7677	**0.8622**	0.8346	**0.8622**
Contraceptive	0.5656	0.5656	0.5679	0.5814	0.5611	0.5566	0.5656	**0.5905**
Penbased	0.9951	0.9613	0.9951	0.9907	0.9822	0.9942	0.9893	**0.9969**
Nursery	0.9133	0.8081	0.8997	0.9270	0.9432	0.8755	0.9380	**0.9884**
Online Shoppers	0.8797	0.8838	0.8786	0.8810	0.8832	0.8829	0.8827	**0.8862**
Adult	0.8483	0.8424	0.8492	0.8494	0.8473	0.7764	0.8549	**0.8594**
Bank Marketing	0.9068	0.9103	0.9073	0.9102	0.9062	0.8256	0.9089	**0.9131**
Connect-4	0.6892	0.6677	0.7061	0.7035	0.6994	0.2339	0.7295	**0.8235**

TABLE V
OPTIMAL HYPERPARAMETERS OF THE MODEL OBTAINED THROUGH GRID SEARCH OR ADAPTIVE ALGORITHMS

Datasets	**FBLS**			**BL-DFIS**	**DFBLS**		**EFBLS**	**T2F-BLS**				**IFBLS**				**ADHFBLS-grid**		**ADHFBLS**	
	N_r	N_t	N_e	N_f	I	R	$[N_r, N_t, N_e]^3$	n	K	m	L	p	m	q	l	R	η	R	m
Diabetes	2	4	32	18	5	4	[5,4,5][1,20,20][1,4,25]	8	4	4	4	2	4	6	16	2	2	7	16
Vehicle	10	4	22	18	3	8	[1,16,25][1,4,10][1,4,10]	10	14	4	16	18	14	2	14	34	4	33	69
Contraceptive	2	5	49	13	5	2	[1,12,30][1,12,25][1,12,5]	8	8	20	4	4	4	12	6	27	1	9	17
Penbased	20	10	100	16	3	20	[5,16,100][1,16,80][5,20,100]	8	20	20	40	25	25	5	5	40	5	257	514
Nursery	20	10	80	16	7	10	[1,12,100][5,12,60][3,4,60]	10	20	20	40	30	30	10	25	39	5	1025	1025
Online Shoppers	18	6	60	17	7	5	[1,20,100][5,20,100][1,8,100]	6	20	4	20	5	5	30	15	34	4	129	131
Adult	20	10	90	14	7	25	[5,20,20][1,12,80][3,16,40]	8	20	12	40	5	25	5	5	37	5	258	260
Bank Marketing	20	10	40	20	7	15	[1,20,40][3,8,80][1,12,60]	6	15	16	40	15	5	5	5	33	5	257	260
Connect-4	20	10	90	42	7	20	[5,20,100][1,20,100][1,8,100]	10	20	20	40	25	30	5	5	39	5	2049	2049

learning, let $\delta \leftarrow 1$. When $\delta = 1$ and there is no improvement in model performance after any incremental learning, we consider the current hyperparameters to be optimal. To prevent randomness, each incremental learning needs to be repeated τ times and select the best performing model, while limiting the maximum search times to t_{max} to prevent unlimited growth of hyperparameters.

The proposed hyperparameter automatic determination algorithm is shown in Algorithm 1, which is also the training algorithm for ADHFBLS. We have the following remarks: 1) The hyperparameter automatic determination algorithm we propose is a heuristic optimization algorithm aimed at quickly adapting to the size of the training dataset. This method can be easily extended to search for multiple variables and be applied to other scenarios where grid search is required to find suitable hyperparameters more quickly; 2) We minimize the impact of randomness by repeatedly testing the model after incremental learning, and this is based on the fact that ADHFBLS has fewer hyperparameters and fast incremental learning for rules and enhancement nodes; 3) The addition of rules inevitably increases the number of enhancement nodes, but the individual increment of enhancement nodes increases the possibility of model performance improvement. Since the number of enhancement nodes is no longer an integer multiple of the number of rules after they are incremented, the algorithm mostly increments the rules before incrementing the enhancement nodes, thus ensuring that there are no more errors.

IV. EXPERIMENTS

A. Experimental Setup

To evaluate the performance of ADHFBLS, we conduct sufficient comparative experiments in this section to verify the performance of the model in all aspects. We first introduce the dataset used in the experiment, which is 9 datasets from UCI [37]. We have selected three small datasets (Diabetes, Vehicle, Contraceptive), three medium datasets (Penbased, Nursery, Online Shoppers), three large datasets (Adult, Bank Marketing, Connect-4), and their detailed information is shown in Table III. We divide the samples into training and testing sets in a 7:3 ratio. To compare the fairness of the experiment, ADHFBLS uses the testing set to test the model's performance during the hyperparameter tuning stage, to find the optimal hyperparameter. Other algorithms that use grid search also use the testing set to find the optimal model. All datasets are normalized and labels are one-hot encoded.

For the setting of ADHFBLS, we set the initial rule number R to 2, the enhancement rate η to be fixed at 1, the number of repetitions τ to 5, and the maximum number of searches t_{max} to 10. These parameters are effective for most datasets, so there is no need to adjust them for datasets of different sizes. Therefore, for each dataset, we have ADHFBLS train and test the data directly without adjusting any hyperparameters. All models are implemented using Python 3.10.12 and tested on servers with Intel (R) Xeon (R) Silver 4316 CPU @ 2.30GHz.

B. Baseline Models

We compare ADHFBLS with some FBLS-based models proposed in recent years to verify its effectiveness in model accuracy and training costs. Since some models based on nonfuzzy and nonBLS have already been compared with these baseline models, we will no longer compare ADHFBLS with these models. The baseline models all need to search for the optimal hyperparameters within the set parameter range through grid search.

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TABLE VI
TOTAL TRAINING TIME OF THE MODELS (TT) AND INDIVIDUAL TRAINING TIME (IT) UNDER OPTIMAL HYPERPARAMETERS

Datasets	FBLS		BL-DFIS		DFBLS		EFBLS		T2F-BLS		IFBLS		ADHFBLS-grid		ADHFBLS
	TT(s)	IT(s)	TT(s)	IT(s)	TT(s)	IT(s)	TT(s)	IT(s)	TT(s)	IT(s)	TT(s)	IT(s)	TT(s)	IT(s)	TT(s)
Diabetes	201.83	0.05	159.46	9.81	90.52	1.72	201.23	0.33	443.47	0.08	214.41	0.04	13.82	0.004	0.68
Vehicle	642.93	0.30	464.59	13.52	118.82	2.16	818.97	1.11	737.35	0.30	197.29	0.02	23.17	0.10	7.88
Contraceptive	1032.39	0.40	466.44	23.82	182.23	2.47	1369.69	2.71	690.00	0.11	283.85	0.01	19.08	0.09	1.09
Penbased	545.09	3.26	686.43	126.61	790.48	36.93	859.43	11.87	376.66	2.70	439.86	0.24	56.85	0.45	104.50
Nursery	604.34	5.14	456.46	157.32	923.43	62.49	832.81	12.43	375.36	3.31	508.68	0.71	52.88	0.35	134.97
Online Shoppers	764.26	3.79	903.34	156.82	949.34	33.65	953.59	19.21	644.33	3.36	482.24	0.20	59.33	0.30	15.60
Adult	1584.99	9.64	2293.31	479.48	3264.93	296.78	2720.08	27.22	1334.86	9.30	2117.44	0.86	151.06	1.20	205.52
Bank Marketing	1959.58	11.77	4207.34	738.60	4001.81	263.24	2941.22	22.85	2306.14	10.20	2374.18	0.33	244.09	1.66	325.78
Connect-4	3448.03	26.91	10023.87	1488.88	5429.99	451.72	5277.14	109.41	8546.65	81.46	3067.83	2.78	328.38	2.50	1651.58

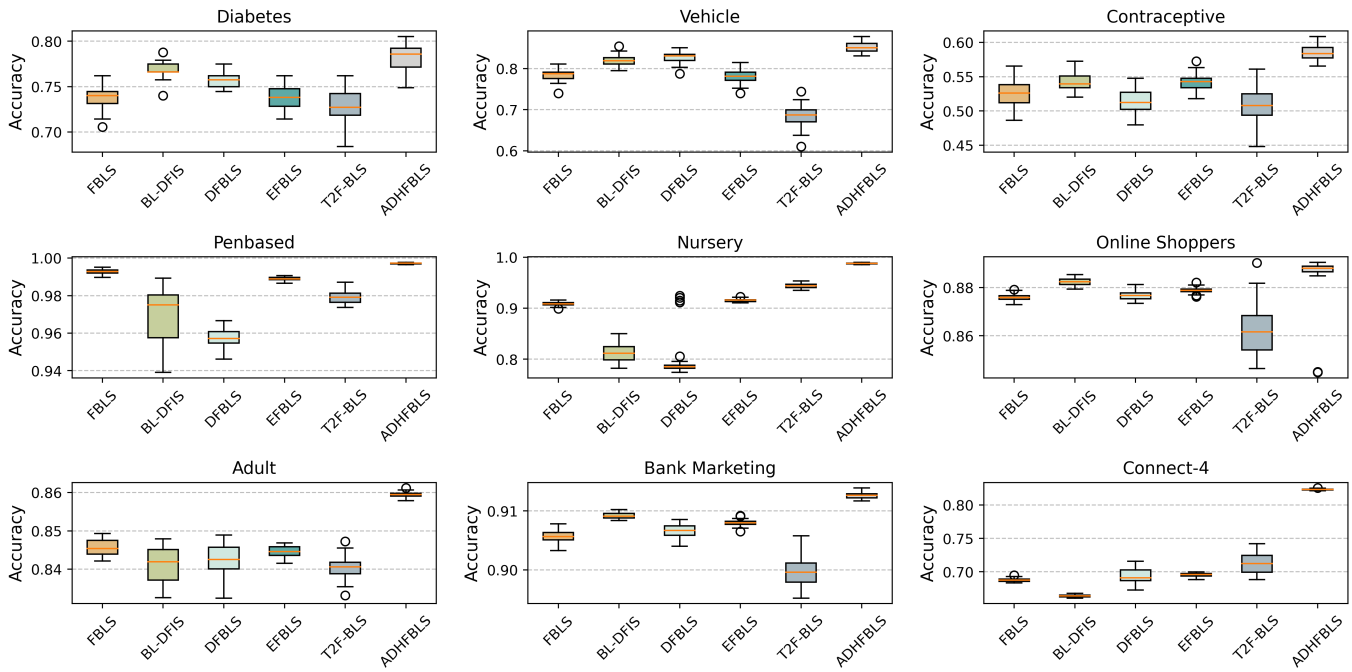

Fig. 3. Box plot of experimental results from multiple model trainings.

BL-DFIS [32] BL-DFIS proposed an adaptive structure for FBLS. The regularization parameter C and the number of feature nodes N_f (No. of features$\times h$) need to be adjusted. Based on the original paper and experimental costs, we fix $C = 10^{-5}$, with $h \in \{0.25, 0.5, ..., 4\}$ in the small datasets and $h \in \{1, 2, 3\}$ in the medium and large datasets.

DFBLS [33] The feature nodes of DFBLS are consistent with FBLS, but the enhancement nodes use the T-S fuzzy model based on ILFRs. We set the regularization parameter $C = 10^{-5}$, the number of subsystems $I \in \{3, 5, 7\}$, the number of rules $R \in \{2, 4, ..., 20\}$ in the small datasets, and $R \in \{5, 10, ..., 30\}$ in the medium and large datasets.

EFBLS [34] EFBLS achieves a residual network through multiple FBLS, further improving accuracy. We set the number of blocks for FBLS to 3 and adjust hyperparameters for each FBLS one by one. In the small datasets, the number of rules $N_r \in \{1, 2..., 5\}$, the number of subsystems $N_t \in \{4, 8, ..., 20\}$, the number of enhancement nodes $N_e \in \{5, 10, ..., 50\}$, and in the medium and large datasets, $N_r \in \{1, 3, 5\}$, $N_t \in \{4, 8, ..., 20\}$, $N_e \in \{20, 40, .., 100\}$.

T2F-BLS [35] T2F-BLS is also based on IT2FS for FBLS, but its proposed incremental learning only focuses on the increment of training data. In the small datasets, the number of subsystems $n \in \{2, 4, ..., 10\}$, the number of rules $K \in \{2, 4, ..., 20\}$, the number of group enhancement neurons $m \in \{4, 8, ..., 20\}$ and the number of neurons $L \in \{4, 8, ..., 40\}$ in a group. In the medium and large datasets, $n \in \{2, 4, ..., 10\}$, $K \in \{5, 10, ..., 20\}$, $m \in \{4, 8, ..., 20\}$, $L \in \{20, 40\}$.

IFBLS [38] IFBLS uses a fuzzy-based approach to weight the error matrix of the BLS to enhance the robustness of the model. The number of feature nodes p, groups m, enhancement nodes q, groups l are all chosen among $\{2, 4, ..., 20\}$ in the small datasets and among $\{5, 10, ..., 30\}$ in the medium and large datasets.

FBLS [21] Similar to EFBLS, $N_r \in \{1, 2, ..., 10\}$, $N_t \in \{1, 2, ..., 5\}$, $N_e \in \{1, 2, ..., 50\}$ in the small datasets and $N_r \in \{4, 8, ..., 20\}$, $N_t \in \{2, 4, ..., 10\}$, $N_e \in \{10, 20, ..., 100\}$ in the medium and large datasets.

ADHFBLS-grid We also compare the performance of ADHFBLS using grid search directly. In any dataset, the number of rules $R \in \{1, 2, ..., 40\}$ and the enhancement rate $\eta \in \{1, 2, ..., 6\}$.

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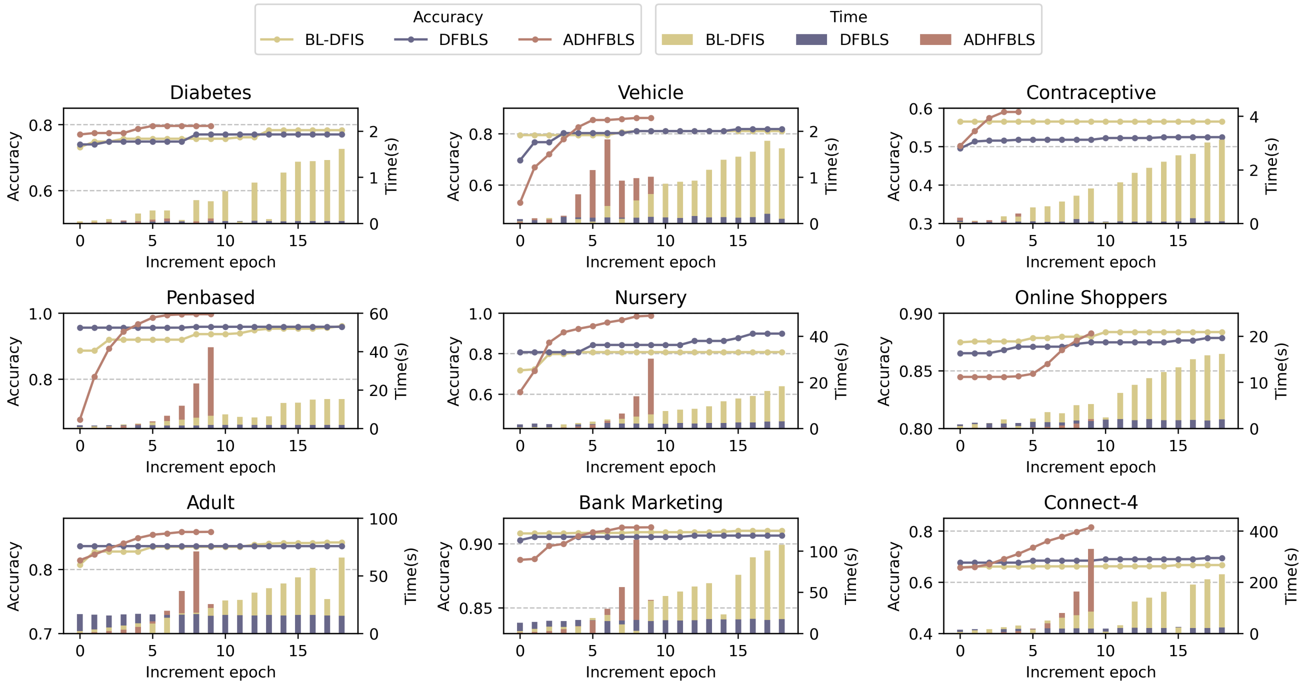

Fig. 4. Accuracy and time consumed per round during incremental learning for ADDHFBLS, BL-DFIS, and DFBLS.

It should be noted that both BL-DFIS and DFBLS have the characteristic of adaptive incremental learning, with the maximum enhancement nodes set to 20, and they will directly use the testing set to guide their increments.

C. Evaluation of Model Accuracy and Training Time

In this section, we set fixed random seeds for the models and then train these models to obtain the final accuracy, optimal hyperparameters, and training time. Accuracy is the ratio of the number of samples correctly classified by the model to the number of all samples in the dataset, and a higher accuracy indicates a more efficient model. The training time of the model consists of a total training time TT using grid search and a single training time IT with fixed optimal hyperparameters. ADHFBLS only has a total training time since it does not require a grid search.

The experimental results are shown in Tables IV, V, and VI. From the experimental results, it can be seen that ADHFBLS has higher accuracy and lower training time compared to other models. Especially on large-scale datasets, the training time of ADHFBLS on the Adult and Bank Marketing datasets is 205.52s and 325.78s, respectively, while other models require more than 1000s. Regardless of the size of the dataset, ADHFBLS can always find hyperparameters that match the size of the dataset through self-adaption, thereby maximizing accuracy in a short time. For the baseline models, although the individual training time of FBLS, EFBLS, T2F-BLS, and IFBLS is very short, it brings a long total training time due to grid search. Moreover, the scope of the grid search still needs to be manually determined, which may be directionless. For example, in Connect-4, the hyperparameters of the ADHFBLS grow to the maximum of the experimental setup to accommodate large-scale data. However, the baseline models are limited by the grid search and do not find suitable hyperparameters, leading to a considerable accuracy gap. The accuracy performance of the ADHFBLS-grid is not as good as other baseline models, indicating that the key to improving ADHFBLS performance lies in the automatic determination of hyperparameters, which is based on the premise that ADHF-BLS has fewer hyperparameters and an efficient incremental learning algorithm.

D. Evaluation of Model Stability

Due to the high randomness of BLS-based models, we eliminate fixed random seeds and train these models multiple times to evaluate the stability of the models. The baseline models are trained using the optimal hyperparameters obtained from the previous experiment, while ADHFBLS is trained from scratch. IFBLS and ADHFBLS-grid are not involved in the experiment, and each model is trained 30 times repeatedly.

The experimental data shown in Fig. 3 indicates that ADHF-BLS maintains high accuracy in multiple runs. The accuracy of ADHFBLS in Fig. 3 partially exceeds that of ADHFBLS in Table IV on each dataset, indicating that the accuracy in Table IV is not a coincidence. The box plot reflects the distribution of model accuracy. The smaller the box length, the more concentrated the distribution is, implying smaller variance and more stable model performance. Fig. 3 demonstrates that AD-HFBLS has good stability, especially on Penbased, Nursery, and Connect-4 datasets. Even with the same parameters, the final accuracy fluctuation of baseline models is significant. Compared with the accuracy of the previous experiment, some models do not achieve high accuracy through grid search, indicating that grid search has flaws because the parameters

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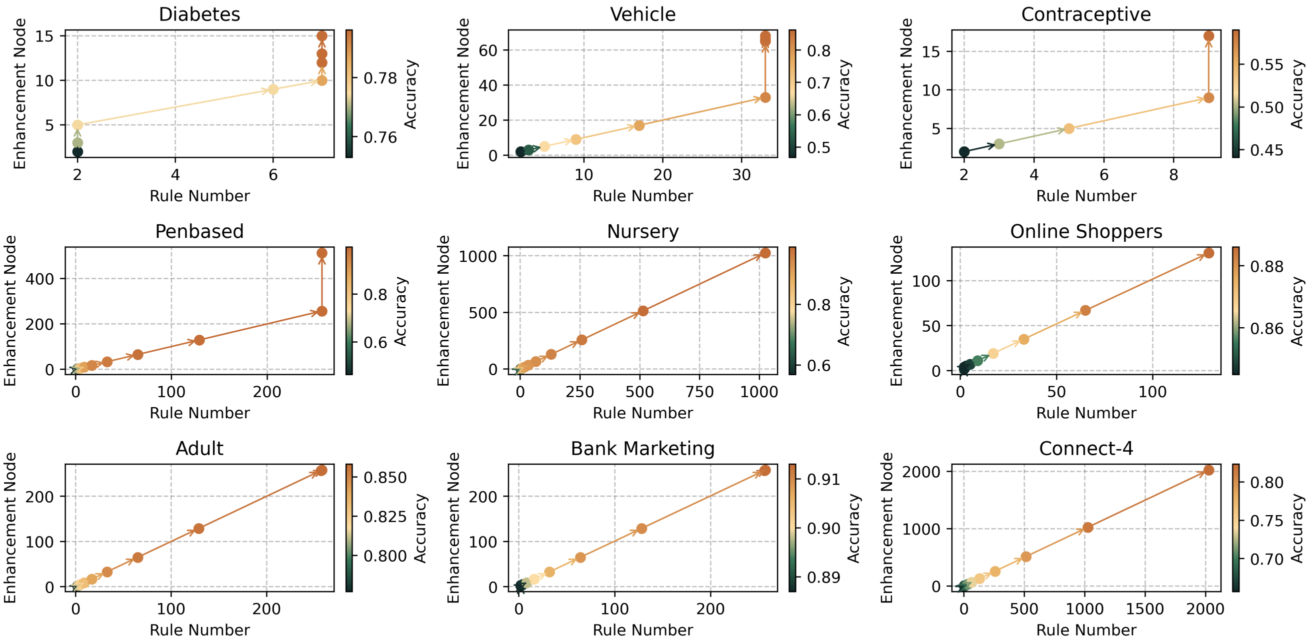

Fig. 5. Changes in the number of rules and enhancement nodes in the automatic determination of hyperparameters for ADHFBLS.

determined through grid search are highly random. In addition, ADHFBLS minimizes the randomness of model performance evaluation by running multiple times when searching for network structures. However, this is based on the fact that ADHFBLS itself searches for network structures quickly. For other models, evaluating a model multiple times during grid search will bring greater time consumption.

E. Evaluation of model adaptability

In this section, we discuss the performance changes of the model during the self-adaption. ADHFBLS, BL-DFIS, and DFBLS all have adaptive learning processes. We study their time and model accuracy improvement in each increment during the incremental learning process. The experimental results are shown in Fig. 4. Due to ADHFBLS searching the network structure from scratch, while other models use the hyperparameters of grid search to determine the base model, the accuracy of ADHFBLS is lower at the initial stage. However, during the search process, the accuracy of ADHFBLS gradually improves and surpasses other models, eventually stabilizing, indicating the correctness and effectiveness of ADHFBLS's incremental learning algorithm. ADHFBLS sets a maximum number of searches to prevent unlimited expansion of the model, while BL-DFIS and DFBLS also set a fixed maximum number of enhancement nodes. Through this approach, the expansion of the model can be stopped early rather than waiting for overfitting to occur, making the model size more controllable. Except for the Contraceptive dataset, the hyperparameter search times of ADHFBLS have all reached 10 times, although the accuracy of the model increases very slowly or even no longer improves in the later stages of the search. Therefore, limiting the maximum number of searches helps to reduce the training cost of the model. ADHFBLS still requires a long time in the later incremental learning, but this time consumption is only required once.

We also demonstrate the rule-incrementing and enhancement nodes-incrementing process of ADHFBLS for the automatic determination of hyperparameters, as shown in Fig. 5. From the adaptive increment process of ADHFBLS, it can be seen that ADHFBLS can search for proper hyperparameters without any hyperparameter settings, regardless of the size of the dataset. For small-scale datasets, ADHFBLS can terminate model amplification before overfitting occurs, or slowly grow the model in an attempt to achieve better performance. When the data can no longer be improved by adding fuzzy rules, ADHFBLS further enhances the model's learning ability by separately adding enhancement nodes. For large-scale datasets, ADHFBLS grows rapidly because the increase in hyperparameters δ, is exponentially increasing. When the hyperparameters are close to being able to match the size of the dataset, δ is reset to 1, and then the hyperparameters are fine-tuned to further match the size of the dataset. Therefore, ADHFBLS has a very flexible structure and an efficient hyperparameter regulation mechanism. The rules are incremented more frequently because they are accompanied by an increment in the enhancement nodes, but the separate increment of the enhancement nodes gives the model more flexibility.

F. Nonparametric Statistical Analysis

TABLE VII
FRIEDMAN TEST FOR MODEL ACCURACY

Models	FBLS	BL-DFIS	DFBLS	EFBLS	T2F-BLS	IFBLS	ADHFBLS
Ranking	4.7222	4.8333	4.1667	3.3333	4.5556	5.2778	1.1111
p-value	6.54×10^{-4}						

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Finally, we conduct the Friedman test and Holm post-hoc test [39] on the ADHFBLS and the baseline model to demonstrate significant differences in model performance. First, the Friedman test is performed with confidence level $\alpha = 0.05$. We rank the accuracy of Table IV and then calculate the average ranking of each model under all datasets and calculate the p-value. As shown in Table VII, p-value=$6.54\times10^{-4} \ll \alpha$. A smaller p-value indicates a more significant performance of the models. Thus, we conclude that there is a significant difference in the performance of these models on the datasets at $\alpha = 0.05$, thus indicating that ADHFBLS has excellent performance, with an average ranking of 1.1111.

TABLE VIII
HOLM TEST FOR MODEL ACCURACY

i	Models	z	p	$Holm$	Hypothesis
6	IFBLS	4.0916	0.0000	0.0083	Rejected
5	BL-DFIS	3.6551	0.0003	0.0100	Rejected
4	FBLS	3.5460	0.0004	0.0125	Rejected
3	T2F-BLS	3.3824	0.0007	0.0167	Rejected
2	DFBLS	3.0005	0.0027	0.0250	Rejected
1	EFBLS	2.1822	0.0291	0.0500	Rejected

Then, we conduct the Holm post-hoc test to verify the significance of the performance difference between ADHFBLS and other models. The experimental results are shown in Table VIII. z indicates the ranking difference between ADHFBLS and the corresponding model, and the larger the z is, the more significant the difference is. p indicates the probability of occurrence of the hypothesis that the difference in performance between ADHFBLS and the corresponding model is not significant. $Holm$ is the adjusted threshold. When $Holm$ is larger than p, it indicates that there is a significant difference between the models. In the experimental results, all hypotheses are rejected, indicating that there is a statistically significant difference between the ADHFBLS and each baseline model. Thus, we have verified through these tests that ADHFBLS has a very significant performance improvement over the existing models, and that this improvement is statistically significant and not due to coincidence.

V. CONCLUSION AND PROSPECT

In this paper, we propose ADHFBLS to address the difficulties in determining hyperparameters in existing FBLS-based models due to randomness. We propose a novel FBLS based on IT2FS, which reduces the number of hyperparameters and enhances the robustness of the model. Then, we implement incremental learning algorithms for ADHFBLS in terms of enhancement nodes and number of rules, enabling the model to quickly scale up. On this basis, we design a heuristic hyperparameter optimization algorithm to quickly find hyperparameters that match the size of the dataset, thereby improving the accuracy of the model and reducing training time. The experimental results fully demonstrate that ADHFBLS has good accuracy, short training time, and a very stable training effect. Therefore, ADHFBLS can be well applied in scenarios with limited computing resources, high real-time requirements, and demand for robustness and interpretability. However, ADHFBLS has only been tested on some publicly available datasets and has not yet been deployed to real-world applications. In future research, we will further investigate how to apply ADHFBLS to practical problems and further optimize ADHFBLS based on actual applications.

REFERENCES

[1] C. P. Chen and Z. Liu, "Broad learning system: An effective and efficient incremental learning system without the need for deep architecture," *IEEE transactions on neural networks and learning systems*, vol. 29, no. 1, pp. 10–24, 2017.

[2] R. Zeng, H. Liu, S. Peng, L. Cao, A. Yang, C. Zong, and G. Zhou, "Cnn-based broad learning for cross-domain emotion classification," *Tsinghua Science and Technology*, vol. 28, no. 2, pp. 360–369, 2023.

[3] X. Wang, T. Huang, K. Zhu, and X. Zhao, "Lstm-based broad learning system for remaining useful life prediction," *Mathematics*, vol. 10, no. 12, p. 2066, 2022.

[4] X. Gong, T. Zhang, C. L. P. Chen, and Z. Liu, "Research review for broad learning system: Algorithms, theory, and applications," *IEEE Transactions on Cybernetics*, vol. 52, no. 9, pp. 8922–8950, 2022.

[5] C. Zhang, S. Ding, L. Guo, and J. Zhang, "Broad learning system based ensemble deep model," *Soft Computing*, vol. 26, no. 15, pp. 7029–7041, 2022.

[6] F. Chu, G. Wang, J. Wang, C. P. Chen, and X. Wang, "Learning broad learning system with controllable sparsity through l0 regularization," *Applied Soft Computing*, vol. 136, p. 110068, 2023.

[7] D. Zhang, C. P. Chen, T. Li, Y. Zuo, and N. Q. Duy, "Target tracking method of siamese networks based on the broad learning system," *CAAI Transactions on Intelligence Technology*, vol. 8, no. 3, pp. 1043–1057, 2023.

[8] J. Yan, M. Yi, X. Yang, X. Luo, and X. Guan, "Broad learning-based localization for underwater sensor networks with stratification compensation," *IEEE Internet of Things Journal*, 2023.

[9] L. Xu, C. P. Chen, F. Qing, X. Meng, Y. Zhao, T. Qi, and T. Miao, "Graph-represented broad learning system for landslide susceptibility mapping in alpine-canyon region," *Remote Sensing*, vol. 14, no. 12, p. 2773, 2022.

[10] J. Lu, G. Ma, and G. Zhang, "Fuzzy machine learning: A comprehensive framework and systematic review," *IEEE Transactions on Fuzzy Systems*, vol. 32, no. 7, pp. 3861–3878, 2024.

[11] M. S. Aslam and Z. Ma, "Output regulation for time–delayed takagi–sugeno fuzzy model with networked control system," *Hacettepe Journal of Mathematics and Statistics*, vol. 52, no. 5, pp. 1282–1302, 2023.

[12] M. S. Aslam, H. Bilal, S. S. Band, and P. Ghasemi, "Modeling of nonlinear supply chain management with lead-times based on takagi-sugeno fuzzy control model," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108131, 2024.

[13] H. Yu, J. Lu, and G. Zhang, "Topology learning-based fuzzy random neural networks for streaming data regression," *IEEE Transactions on Fuzzy Systems*, vol. 30, no. 2, pp. 412–425, 2022.

[14] A. K. De, D. Chakraborty, and A. Biswas, "Literature review on type-2 fuzzy set theory," *Soft Computing*, vol. 26, no. 18, pp. 9049–9068, 2022.

[15] R.-E. Precup, R.-C. David, R.-C. Roman, A.-I. Szedlak-Stinean, and E. M. Petriu, "Optimal tuning of interval type-2 fuzzy controllers for nonlinear servo systems using slime mould algorithm," *International Journal of Systems Science*, vol. 54, no. 15, pp. 2941–2956, 2023.

[16] A. Al-Mahturi, F. Santoso, M. A. Garratt, and S. G. Anavatti, "A novel evolving type-2 fuzzy system for controlling a mobile robot under large uncertainties," *Robotics*, vol. 12, no. 2, p. 40, 2023.

[17] B. Meniz and E. M. Özkan, "Vaccine selection for covid-19 by ahp and novel vikor hybrid approach with interval type-2 fuzzy sets," *Engineering applications of artificial intelligence*, vol. 119, p. 105812, 2023.

[18] Z. Liu, C. Wu, X. Shen, W. Yao, J. Liu, and L. Wu, "Adaptive interval type-2 fuzzy neural network-based novel fixed-time backstepping control for uncertain euler–lagrange systems," *IEEE Transactions on Fuzzy Systems*, vol. 32, no. 5, pp. 2966–2975, 2024.

[19] M. Tabakov, A. B. Chlopowiec, and A. R. Chlopowiec, "A novel classification method using the takagi–sugeno model and a type-2 fuzzy rule induction approach," *Applied Sciences*, vol. 13, no. 9, p. 5279, 2023.

[20] H. Zhou, H.-K. Lam, B. Xiao, and C. Xuan, "Integrated fault-tolerant control design with sampled-output measurements for interval type-2 takagi–sugeno fuzzy systems," *IEEE Transactions on Cybernetics*, 2024.

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[21] S. Feng and C. P. Chen, "Fuzzy broad learning system: A novel neuro-fuzzy model for regression and classification," *IEEE transactions on cybernetics*, vol. 50, no. 2, pp. 414–424, 2018.

[22] C. Peng and D. ChunHao, "Monitoring multi-domain batch process state based on fuzzy broad learning system," *Expert Systems with Applications*, vol. 187, p. 115851, 2022.

[23] J. Huang, C.-M. Vong, G. Wang, W. Qian, Y. Zhou, and C. P. Chen, "Joint label enhancement and label distribution learning via stacked graph regularization-based polynomial fuzzy broad learning system," *IEEE Transactions on Fuzzy Systems*, 2023.

[24] H.-G. Han, F.-F. Yang, H.-Y. Yang, and X.-L. Wu, "Type-2 fuzzy broad learning controller for wastewater treatment process," *Neurocomputing*, vol. 459, pp. 188–200, 2021.

[25] X. Wang, C. Wang, K. Zhu, and X. Zhao, "A mechanical equipment fault diagnosis model based on tsk fuzzy broad learning system," *Symmetry*, vol. 15, no. 1, p. 83, 2022.

[26] C.-C. Tsai, H.-Y. Chen, C.-C. Chan, C.-C. Hung, and G.-M. Chen, "Intelligent actor-critic learning control for collison-free trajectory tracking of mecanum-wheeled mobile robots," *International Journal of Fuzzy Systems*, vol. 26, no. 4, pp. 1133–1142, 2024.

[27] X. Hu, G. Zhao, A. Dong, G. Lv, Y. Zhai, Y. Guo, and X. Dong, "Few-shot hyperspectral image classification with spectral-spatial feature fusion based on fuzzy broad learning system," in *2023 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2023, pp. 3160–3164.

[28] Z. Huang and J. Duan, "Gfbls: Graph-regularized fuzzy broad learning system for detection of interictal epileptic discharges," *Engineering Applications of Artificial Intelligence*, vol. 125, p. 106763, 2023.

[29] S. Feng, C. L. P. Chen, L. Xu, and Z. Liu, "On the accuracy–complexity tradeoff of fuzzy broad learning system," *IEEE Transactions on Fuzzy Systems*, vol. 29, no. 10, pp. 2963–2974, 2021.

[30] W. Zou, Y. Xia, and L. Dai, "Fuzzy broad learning system based on accelerating amount," *IEEE Transactions on Fuzzy Systems*, vol. 30, no. 9, pp. 4017–4024, 2022.

[31] M. Jiang, W. Zhang, and H. Zhang, "Broad learning system based on triangular fuzzy membership function," in *2023 8th International Conference on Image, Vision and Computing (ICIVC)*, 2023, pp. 842–846.

[32] K. Bai, X. Zhu, S. Wen, R. Zhang, and W. Zhang, "Broad learning based dynamic fuzzy inference system with adaptive structure and interpretable fuzzy rules," *IEEE Transactions on Fuzzy Systems*, vol. 30, no. 8, pp. 3270–3283, 2022.

[33] H. Zhao, Y. Wu, and W. Deng, "An interpretable dynamic inference system based on fuzzy broad learning," *IEEE Transactions on Instrumentation and Measurement*, 2023.

[34] J. Duan, S. Yao, J. Tan, Y. Liu, L. Chen, Z. Zhang, and C. P. Chen, "Extreme fuzzy broad learning system: Algorithm, frequency principle, and applications in classification and regression," *IEEE Transactions on Neural Networks and Learning Systems*, 2024.

[35] H. Han, Z. Liu, H. Liu, J. Qiao, and C. L. P. Chen, "Type-2 fuzzy broad learning system," *IEEE Transactions on Cybernetics*, vol. 52, no. 10, pp. 10 352–10 363, 2022.

[36] C. Sun, H. Han, X. Wu, and H. Yang, "Antiforgetting incremental learning algorithm for interval type-2 fuzzy neural network," *IEEE Transactions on Fuzzy Systems*, vol. 32, no. 4, pp. 1938–1950, 2024.

[37] C. Blake, "Uci repository of machine learning databases," http://www.ics.uci.edu/~mlearn/MLRepository.html, 1998, [Online]. Available.

[38] M. Sajid, A. K. Malik, and M. Tanveer, "Intuitionistic fuzzy broad learning system: Enhancing robustness against noise and outliers," *IEEE Transactions on Fuzzy Systems*, vol. 32, no. 8, pp. 4460–4469, 2024.

[39] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *The Journal of Machine learning research*, vol. 7, pp. 1–30, 2006.

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